

Detailed Classification Reports

◆ Random Forest

	precision	recall	f1-score	support
Normal	0.82	0.73	0.77	88
Glioma Tumor	0.82	0.77	0.79	180
Meningioma Tumor	0.69	0.73	0.71	183
Pituitary Tumor	0.82	0.87	0.84	169
accuracy			0.78	620
macro avg	0.79	0.77	0.78	620
weighted avg	0.78	0.78	0.78	620

◆ Logistic Regression

	precision	recall	f1-score	support
Normal	0.83	0.65	0.73	88
Glioma Tumor	0.70	0.74	0.72	180
Meningioma Tumor	0.72	0.74	0.73	183
Pituitary Tumor	0.85	0.87	0.86	169
accuracy			0.76	620
macro avg	0.77	0.75	0.76	620
weighted avg	0.77	0.76	0.76	620

◆ SVC

	precision	recall	f1-score	support
Normal	0.86	0.76	0.81	88
Glioma Tumor	0.77	0.77	0.77	180
Meningioma Tumor	0.74	0.79	0.77	183
Pituitary Tumor	0.86	0.85	0.85	169
accuracy			0.80	620
macro avg	0.81	0.79	0.80	620
weighted avg	0.80	0.80	0.80	620

◆ Decision Tree

	precision	recall	f1-score	support
Normal	0.64	0.66	0.65	88
Glioma Tumor	0.61	0.63	0.62	180
Meningioma Tumor	0.58	0.61	0.60	183
Pituitary Tumor	0.67	0.60	0.63	169
accuracy			0.62	620
macro avg	0.62	0.62	0.62	620
weighted avg	0.62	0.62	0.62	620

◇ KNN

	precision	recall	f1-score	support
Normal	0.68	0.70	0.69	88
Glioma Tumor	0.60	0.87	0.71	180
Meningioma Tumor	0.84	0.51	0.63	183
Pituitary Tumor	0.88	0.81	0.85	169
accuracy			0.72	620
macro avg	0.75	0.72	0.72	620
weighted avg	0.76	0.72	0.72	620

◇ Ensemble (Soft Vote)

	precision	recall	f1-score	support
Normal	0.79	0.80	0.79	88
Glioma Tumor	0.80	0.81	0.80	180
Meningioma Tumor	0.83	0.79	0.81	183
Pituitary Tumor	0.86	0.89	0.88	169
accuracy			0.82	620
macro avg	0.82	0.82	0.82	620
weighted avg	0.82	0.82	0.82	620

📄 MODEL PERFORMANCE SUMMARY

Model	Acc	Prec	Rec	F1	Spec	AUC
Random Forest	0.779	0.787	0.774	0.779	0.924	0.940
Logistic Regression	0.763	0.775	0.750	0.759	0.918	0.908
SVC	0.795	0.807	0.792	0.798	0.929	0.952
Decision Tree	0.619	0.624	0.624	0.624	0.870	0.747
KNN	0.724	0.750	0.724	0.720	0.906	0.917
Ensemble (Soft Vote)	0.824	0.820	0.822	0.821	0.940	0.952

🏆 Best Model (by AUC): Ensemble (Soft Vote)

✅ Ensemble (Soft Vote) model saved to 'brain_tumor_best_model.pkl'

📊 Ensemble vs Individual Models:

Model	AUC
Random Forest	0.940
Logistic Regression	0.908
SVC	0.952
Decision Tree	0.747
KNN	0.917
Ensemble (Soft Vote)	0.952

🏆 Final model for prediction: Ensemble (Soft Vote)